# ABSTRACT

The death rate from lung cancer has increased in recent years, it is essential to determine whether the tumor has progressed to cancer. if the prediction is made early, many lives can be saved, and it is an accurate prediction. can assist the physicians in beginning their treatment as well. computed tomography is essential for determining the state of the tumor by examining its size, location, and other characteristics.In order to save many lives that are in danger, we have put forth a framework in this paper for early cancer prediction.in essence, we concentrate on two areas of computer science: machine learning and digital image processing, or dip. the preprocessing stage of digital image processing is widely recognized. at a later stage, the previously processed image is exposed to a segmentation phase, after which the segmented image is subjected to feature extraction.lastly, machine learning classification algorithms such as random forest, svc (support vector classification), and histgradient boosting cassifier(hgbc)are used to train the extracted features. it is predicted whether the tumor is benign or malignant based on the classification results that were obtained. to find the algorithm with the highest predictive accuracy, the necessary parameters—accuracy, recall, and precision—are computed.

# CHAPTER 1

# INTRODUCTION

## 1.1 OVERVIEW

Machine Learning is the field of study that gives computers the capability to learn without being explicitly programmed. ML is one of the most exciting technologies that one would have ever come across. As it is evident from the name, it gives the computer that makes it more similar to humans: The ability to learn. Machine learning is actively being used today, perhaps in many more places than one would expect.

Machine learning, as a powerful approach to achieve Artificial Intelligence, has been widely used in pattern recognition, a very basic skill for humans but a challenge for machines. Nowadays, with the development of computer technology, pattern recognition has become an essential and important technique in the field of Artificial Intelligence. The pattern recognition can identify letters, images, voice or other objects and also can identify status, extent or other abstractions.

The computer was in vented, it has begun to affect our daily life. It improves the quality of our lives; it makes our life more convenient and more efficient. A fascinating idea is to let a computer think and learn as a human. Basically, machine learning is to let a computer develop learning skills by itself with given knowledge. Pattern recognition can be treated like computer being able to recognize different species of objects. Therefore, machine learning has close connection with pattern recognition.

Machine Learning is a scientific research of statistical procedures and methods which they are used by computer systems designed to perform such functions without specific instructions, rather than trusting in the models and conclusions. This is believed to be part of an artificial intelligence. Machine Learning algorithms sets up a mathematical model based on data examples called "training data" to make predictions without the completion of a task being explicitly programmed.

## 1.2 LUNG TUMOR

The cause of lung cancer is aberrant cell division in the lung, which results in abnormal lung function. As of right now, over 100 different forms of lung cancer have been identified. New cells typically replace aging cells in the body. When tumors occur, the cells proliferate quickly and are not replaced by new ones. This will cause cells to accumulate and create

masses known as tumors.any part of the human lung can develop lung cancer, and the symptoms are specific to the area of the lung that is impacted.

### 1.2.1 Signs of Lung Cancer

The following lists the main signs of lung tumors:

- ✓ Respiratory: shortness of breath, wheezing, coughing up blood, or mild coughing
- ✓ Systemic: weakness, fever, weight loss, or clubbed fingernails
- ✓ Pain in the chest, bone, superior obstruction at the venacava, or trouble swallowing are signs that the cancer is pressing on nearby structures.

### 1.2.2 Lung Cancer Grading

To improve communication, treatment planning, and outcome prediction, lung cancers are graded. Tumor severity is indicated by the grade, which goes from I to IV. Grade I tumors are easily treated and grow slowly. However, grade IV tumors are aggressive, rapidly expanding, malignant tumors that are challenging to cure. The highest grade cell determines the tumor grade, though a single tumor may contain cells of various grades.

The majority of grade I tumors are not cancerous, and they have a very high survival rate. Under a microscope, the cells display their typical characteristics. These tumors have a distinct shape and are growing slowly. The best and most efficient way to treat gradeI tumors is through surgery.

The lung survival rate for grade II is moderate. The cell has somewhat odd characteristics when viewed under a microscope. These tissues have the capacity to proliferate and can spread to neighboring tissues. It can alter its characteristics and grow into a higher-grade tumor.

Abnormal cellular characteristics and the capacity for active reproduction are features of grade III lung cancer. These cells can proliferate and spread to become grade IV tumors, and they grow close to healthy lung tissues. The only effective treatments are chemotherapy and surgery, and survival rates are extremely low. There is a high chance of recurrence following treatment.

The most malignant form of lung cancer is grade IV tumors. Their cellular appearance is asymmetrical. They can impact nearby normal cells and grow quickly.For these cells to be nourished and grow quickly, new blood vessels are created. Lung Table 1.1 displays the WHO tumor grading system. Some tumors lack a specific shape or size and have dead cell areas in the middle.

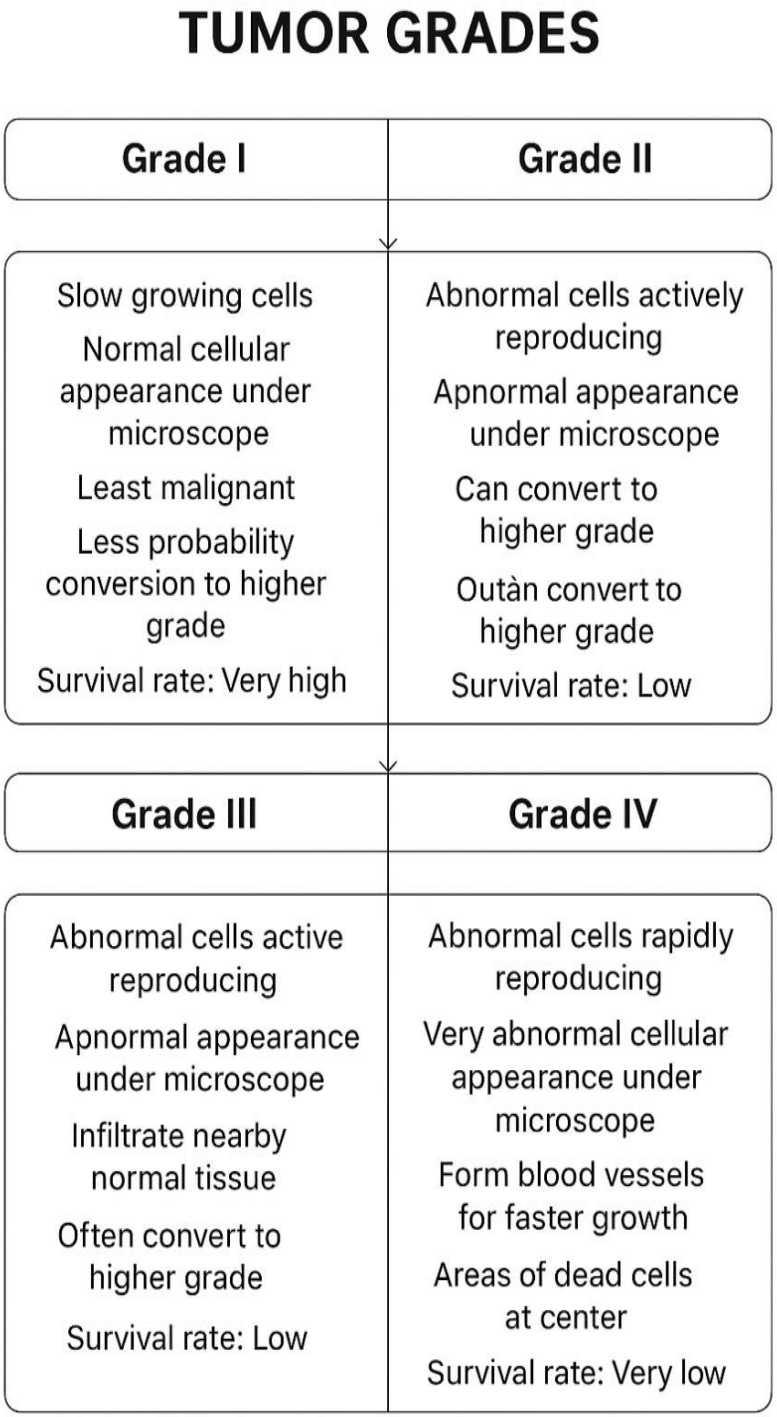


**Figure 1: Lung Tumor Grading System by WHO**

### 1.2.3 Types of Lung Cancer

There are two main types of lung cancer:

They are Non-Small Cell Lung Cancer (NSCLC) and Small Cell Lung Cancer (SCLC). Different forms of growth are found in these two types. They also spread differently hence they are treated in different methods.

**a. Non-Small Cell Lung Cancer**

The non small cell accounts for about 85-90%. This cancer is classified into three and those types are being classified based on the structure of cancer at lungs which depends on the origin and the way it looks. These cells show very slight changes among them. But it generally tries to get almost similar look called as prognosis which undergoes same way of treatement:

**Adenocarcinoma:**

Under NSCLC this is common form of it. In nonsmokers the most commonly found lung cancer is said to be Adenocarcinoma. But it's common with both smokers and former smokers. It actually grows in the lungs outer edges. It has slow growth when compared to that of other lung cancer types.

**Squamous Cell Carcinoma (Epidermoid Carcinoma):**

This type of NSCLC is found more often in smokers or former smokers. It actually grows in the middle part of the lungs near the main airways (the bronchi). Large Cell Carcinoma: This is most rarest type of NSCLC. But it seems dangerous because this type quickly grows and spreads to other organs. Thus it is clear that it is harder to treat such cancer because of its quick growth.

**b. Small Cell Lung Cancer**

SCLC is found in about people at 10% to 15% who are suffering from lung cancer. It is also known to be oat cell cancer. The spreading occurs more quickly when compared to non-small cell lung cancer. At an early stage itself it spreads to other parts of the body.

## 1.3 OVERVIEW OF HUMAN LUNG ANATOMY

The respiratory system of human beings lung is the most important organ. The lungs characterize sponge-like organs in the chest which works effectively as a segment of the respiratory mechanism. The right lung contains three lobes and the left lung is smaller in size with two lobes so as to furnish space for the heart. Air goes through the nose or the mouth and thereafter proceeds through the trachea and each bronchus before passing through the lungs.

The lungs are a couple of sponge- such as cone-shaped organs.Oxygen is full into the lungs when the air is inhaled. Lung tissue forwards oxygen to the bloodstream on its way to the remaining organs of the body. Cells discharge carbon dioxide as they accept oxygen. The bloodstream transports carbon dioxide back to the lungs, and the carbon dioxide makes an exit from the body when the air is exhaled.

The lung, the site of gas exchange, is filled with air that has low density (about1000 HU) on CT images. In addition to air, pulmonary vessels and bronchi are the major elements of the lung regions. Lung regions consist of the left and the right lungs. The left lung is again classified into two lung lobes (upper lobe and lower lobe) by an oblique fissure. The right part of the lung is divided by oblique and horizontal fissures into three lobes (upper lobe, middle lobe, and lower lobe).

The geometry of the bronchial airways in the human chest can be approximately described as a binary tree structure. The trachea (the root of the airway tree) divides into two main branches (left primary bronchi and right primary bronchi), which enter the right and the left lungs. The primary bronchi are divided into five lobar bronchi (two left, three rights) that enter each lung lobe respectively. In each lung region, lobar bronchi divide again and again and make 8–10 segmental bronchi trees.

Over a million valuable lives are lost to lung cancer each year, making it one of the most deadly types of the disease. Thus, lung nodule identification in the chest In today's clinical environment, computed tomography (CT) images are becoming increasingly important. The structure of lung cancer is depicted in Figure 1. 2. The stage at which the disease is diagnosed typically determines the course of treatment and the patients' chances of survival. Therefore, early disease detection is essential to patients' recovery. Early detection of lung cancer is greatly aided by computer-aided diagnosis (CAD) systems.

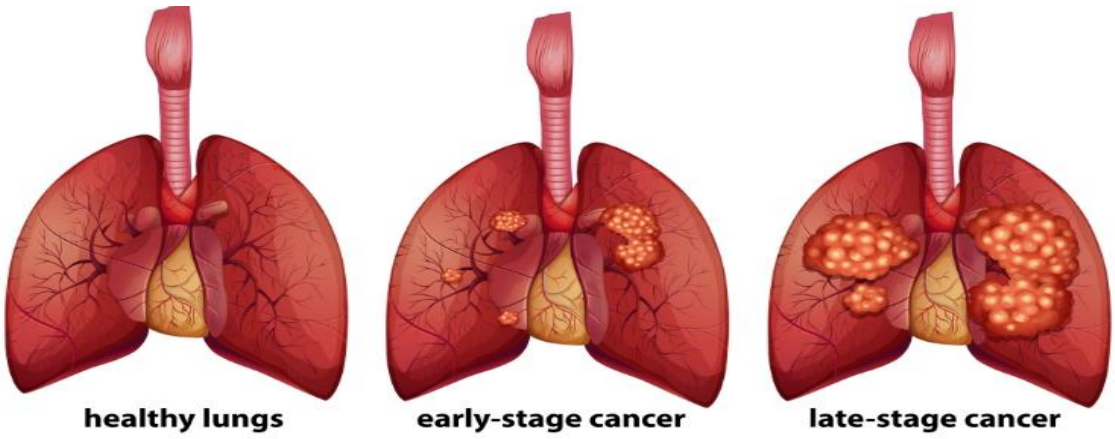


**Figure 2: Normal Lungs and Abnormal Lungs**

### 1.3.1 Symptoms of Lung Tumor

**Hacking cough**

A hacking cough is a persistent, often dry cough that worsens over time as the tumor irritates the airways. This irritation triggers the cough reflex, causing frequent and sometimes spasmodic coughing. As the tumor grows, the cough may become more severe and produce mucus or even blood. This type of cough is common in lung cancer and signals ongoing airway irritation.

**Shortness of breath**

Tumors can block air passages, making it difficult for air to flow freely through the lungs. Additionally, fluid can accumulate around the lungs (pleural effusion), which compresses the lung and prevents full expansion during breathing. These conditions cause shortness of breath by making it harder to breathe deeply and easily. This symptom is common in lung cancer and can significantly impact breathing comfort

**Loss of appetite**

Cancer-related inflammation and the body's stress response can lower appetite quickly and unexpectedly. Tumors release substances that interfere with hunger signals, while the immune response increases inflammation, further suppressing appetite. Additionally, treatments and symptoms like nausea, mouth sores, and fatigue contribute to reduced desire to eat.

**Bleeding**

Tumor growth in the lung can damage blood vessels, leading to bleeding within the lung tissues. This damage causes the person to cough up blood or blood-streaked sputum, a condition known as hemoptysis. The bleeding often originates from bronchial arteries affected by the tumor, and it can range from minor to severe, sometimes becoming life-threatening. Hemoptysis is a common symptom in lung cancer and indicates tumor invasion into blood vessels

**Lump in the neck**

Cancer can spread to lymph nodes, causing them to swell and form lumps. These lumps can be felt in the neck or above the collarbone. Swollen lymph nodes from cancer are usually hard and not painful. If these lumps are noticed, it’s important to see a doctor for evaluation.

**Chest pain**

If the tumor grows into the chest wall, lining, or nerves, it can cause pain. This pain is often felt during coughing, laughing, or taking deep breaths. The discomfort may range from mild to severe, depending on the extent of invasion. Such pain is a common symptom in advanced lung cancer cases and needs medical attention.

**Right-sided abdominal pain**

Lung cancer that spreads to the liver or lymph nodes in the belly can cause pain on the right side of the belly. This pain happens because the cancer affects these organs or tissues. If this pain occurs, it's important to tell a doctor for proper care.

**Blood clots all over the body**

Some cancers produce substances that make blood clot more easily. This can cause clots (thrombosis) to form in different parts of the body. These clots may lead to complications and need medical attention.

**Generalized weakness**

Cancer and its symptoms can cause ongoing tiredness, weakness, and low energy throughout the day. This fatigue is different from normal tiredness because it doesn't go away with rest. It often affects daily activities and overall strength.

**Weight loss**

Unexplained, rapid weight loss in cancer happens because tumors use up a lot of the body's energy, which increases calorie burn even at rest. Cancer can change metabolism and release substances that break down muscle and fat, leading to a loss of both. Even if a person eats less due to reduced appetite or treatment side effects, the body continues to use energy quickly. This process can cause significant weight loss, known as cachexia, and it often occurs alongside other symptoms like fatigue or weakness

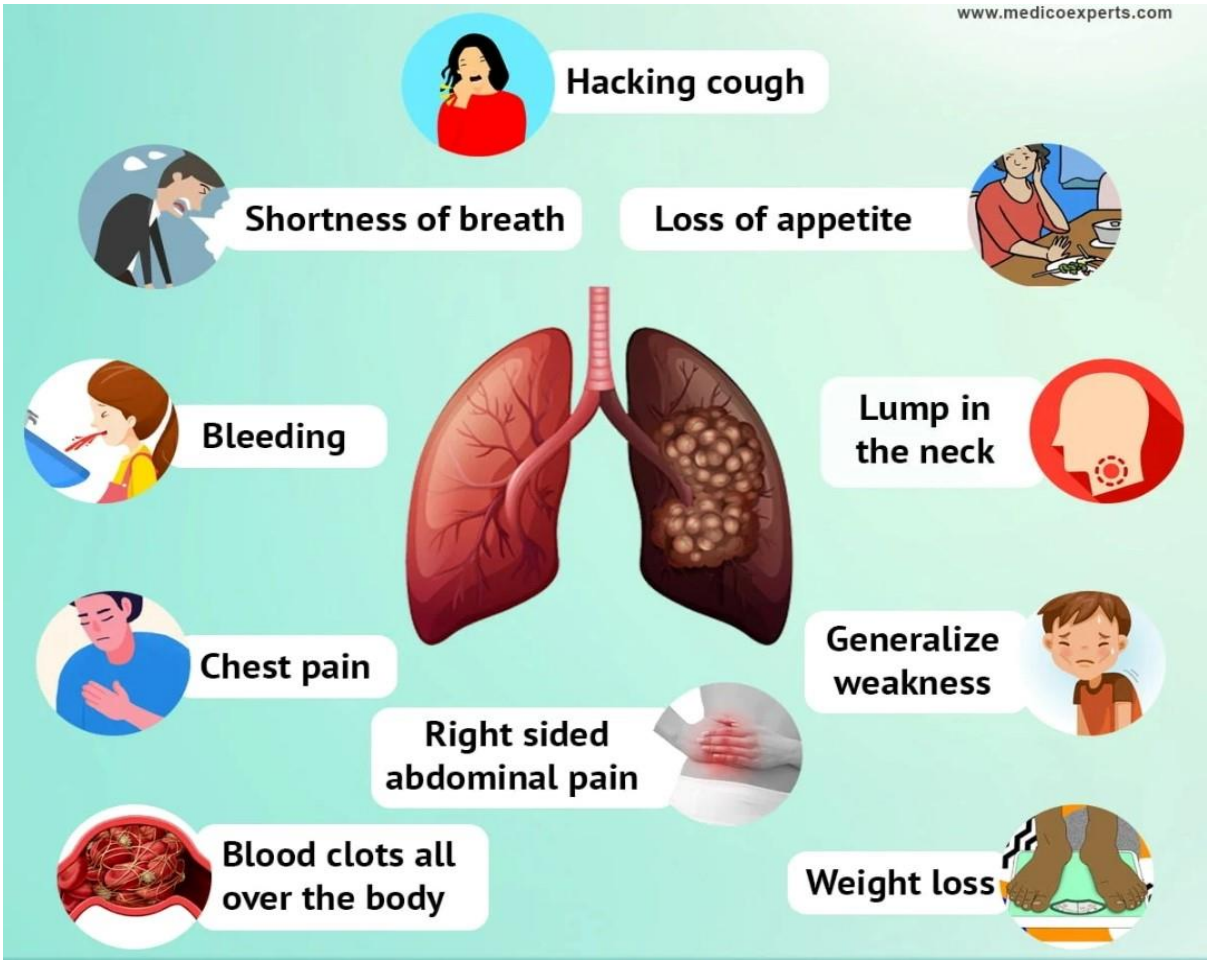


**Figure 3: Symptoms of Lung Tumor**

## 1.4 FACTORS AFFECTING LUNG CANCER

### 1.4.1 Smoking:

Smoking is the leading cause of lung cancer, responsible for nearly 90% of lung cancer deaths. Cigarette smoke contains many carcinogens that damage the DNA in lung cells, causing permanent mutations. These mutations can disrupt normal cell growth controls, leading to uncontrolled cell division and cancer formation. Some chemicals in tobacco smoke also promote inflammation and accelerate the progression of existing lung cancer.

### 1.4.2 Stress (Metal Stress):

Stress from exposure to toxic metals like arsenic, cadmium, chromium, and nickel can significantly increase the risk of lung cancer. These metals cause oxidative stress, damage DNA, and can lead to genetic mutations that disrupt normal cell processes. Chronic metal exposure often deactivates tumor suppressor genes and enhances oncogene activity, allowing abnormal cell growth. The accumulation of these metals in lung tissue over time is linked to higher rates of malignancy and cancer development.

### 1.4.3 Oncogenes:

Oncogenes are genes that, when mutated or overexpressed, drive abnormal cell growth and division in the lungs, leading to cancer. Key oncogenes involved in lung cancer include EGFR, KRAS, and MET, whose activation can disrupt normal cell signaling pathways and promote tumor formation. These changes often result from gene mutations, amplifications, or chromosomal rearrangements. Targeting these oncogenic drivers with specific therapies has become an important approach for treating certain types of lung cancer.

### 1.4.4 Junk Food:

Eating junk food and ultra-processed foods is linked to a significantly higher risk of lung cancer. These foods are often high in unhealthy fats, sugar, and chemical additives that promote inflammation and metabolic changes, which can support cancer development in lung tissue. Studies show that consuming the most ultra-processed foods raises lung cancer risk by up to 41%, even after accounting for other factors like smoking.

### 1.4.5 Harmful Radiation:

Harmful radiation, such as from previous radiation therapy to the chest or exposure to radon gas, increases the risk of lung cancer by damaging lung cells and inducing mutations. This risk is higher in smokers or those with prolonged exposure to radiation. Radiation risk is also influenced by the dose and the lung volume affected during exposure.

### 1.4.6 Harmful Chemicals and Pollution:

Harmful chemicals and pollution increase lung cancer risk by causing DNA damage, chronic inflammation, and oxidative stress in lung tissues. Major contributors include industrial and vehicle emissions, asbestos, arsenic, pesticides, diesel fumes, and indoor pollutants like cooking fuels. Long-term exposure to these pollutants, especially in urban or industrial areas, significantly raises the likelihood of lung cancer

### 1.4.7 Infections:

Infections can increase lung cancer risk by causing chronic inflammation and damage to lung tissues, creating an environment that promotes cancer development. Pulmonary tuberculosis and other chronic lung infections have been linked to a higher risk of lung cancer due to ongoing tissue injury and immune response disruption. Lung cancer patients with infections like tuberculosis also tend to have poorer outcomes.

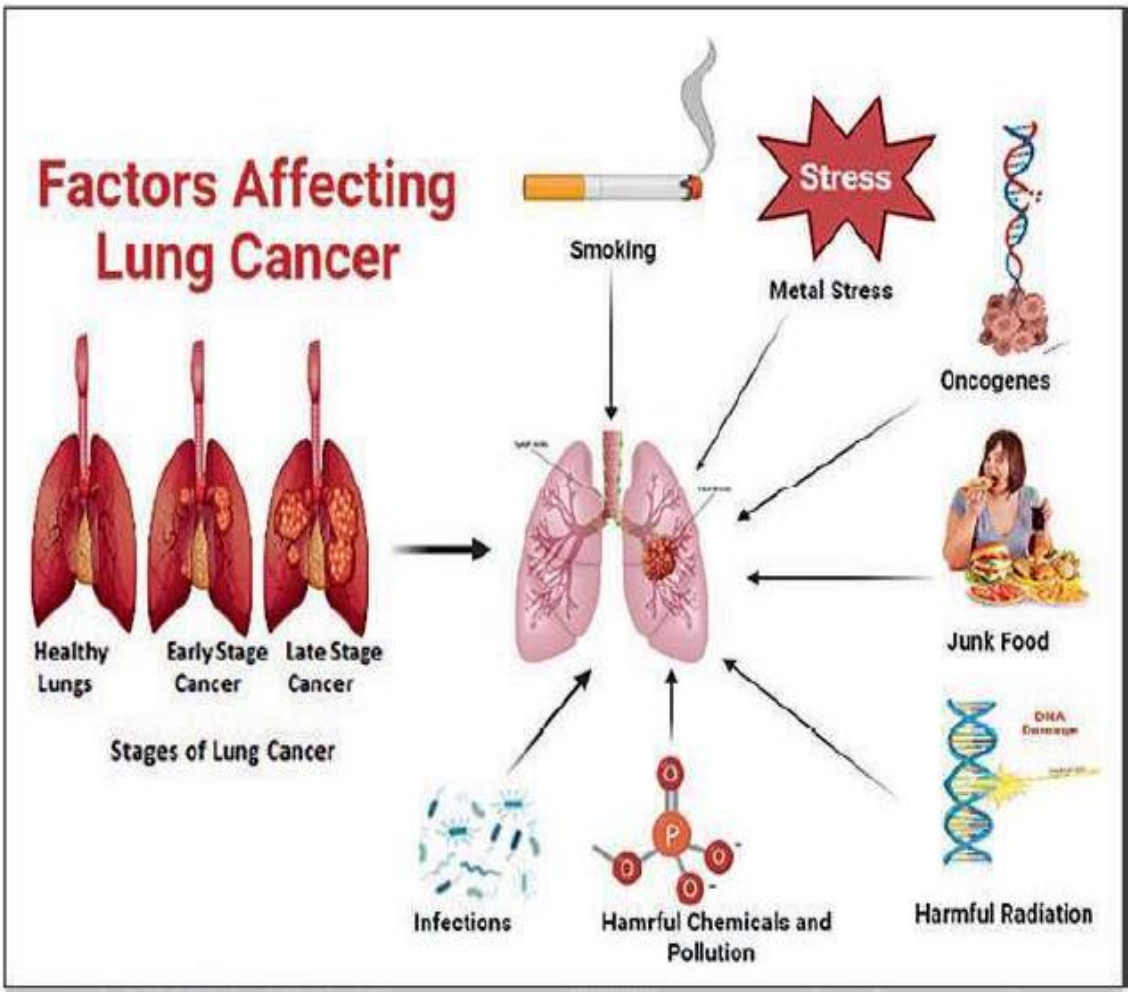


**Figure 4: Factors Affecting Lung Cancer**

### 1.5 TREATMENT

Treatments for lung cancer are offered based on the patient's health, location, and stage. Despite the fact that there are other options, surgery and radiation therapy are the most often recommended methods for curing lung cancer. Physicians use chemotherapy to treat lung cancer.

#### 1.5.1 Surgery

Doctors can perform surgery to remove cancerous lung tissue that has spread to other locations. A lobe or significant portion of the lung is removed during a lobectomy. There is a requirement for removal once it reaches a certain level of severity. Following lung Prior to surgery, appropriate health conditions must be maintained.

### 1.5.2 Chemotherapy

In order to eradicate cancer cells, drugs are an essential part of this treatment. Chemotherapy drugs are used to treat cancer by causing the cells to divide quickly, which produces ideality. Despite being an intervention, chemotherapy can also result in fewer They have a major effect on cancers that have spread to different parts of the body and are attacking the entire body, causing side effects like nausea and weight loss.

### 1.5.3 Radiation Therapy

This method uses high-energy rays to destroy cancerous cells. A tumor can be surgically removed after being shrunk by a doctor using radiation. Radiation therapy may be used if the cancer started in one place and did not spread to other places. is mostly favored.

## 1.6 TARGETED THERAPY

Behavior of cancer cells is specifically target by the use of particular medications. Examples medicines that eradicate cancer from getting multiplied.Medical professionals work together in the areas being treated during such treatment; these specialists could include:

- Surgeons
- Radiation Oncologists
- Specialists in Lung Treatment called Pulmonologists
- Pulmonary Therapists

### 1.6.1 Tumor Diagnosis

If a lesion is suspicious on a lung cancer which is got by screening or experiencing lung cancer symptoms is being identified by doctors, then the next step is confirmed by several diagnostic tests that are available.

### 1.6.2 Imaging Tests

Imaging scans are essential to the treatment of patients with non-small cell lung cancer. Since no other test is perfect and no scan can identify NSCLC, only a biopsy can do so. The results of a person's chest scan and x-ray must be compared with their medical history. In Physical examinations are necessary in order to determine when it started, and data will be gathered from the biopsy.

## 1.7 PURPOSE OF THE MACHINE LEARNING

Machine learning is an application of artificial intelligence (AI) that provides systems the ability to automatically learn and improve from experience without being explicitly programmed. Machine learning focuses on the development of computer programs that can access data and use it learn for themselves.

Nowadays, with the development of computer technology, pattern recognition has become an essential and important technique in the field of Artificial Intelligence. The pattern recognition can identify letters, images, voice or other objects and also can identify status, extent or other abstractions.

The process of learning begins with observations or data, such as examples, direct experience, or instruction, in order to look for patterns in data and make better decisions in the future based on the examples that we provide. The primary aim is to allow the computers learn automatically without human intervention or assistance and adjust actions accordingly.

## 1.8 PROBLEM STATEMENT

The rapid increase in population rate, the rate of diseases like cancer, chikungunya, cholera etc., are also increasing. Among all of them, cancer is becoming a common cause of death. Cancer can start almost anywhere in the human body, which is made up of trillions of cells. Normally, human cells grow and divide to form new cells as the body needs them. When cells grow older or become damaged, they die, and new cells take their place. When cancer cells develop, however, this orderly process breaks down.

As cells become more and more abnormal, old or damaged cells survive when they should die, and new cells form when they are not needed. These extra cells can divide without stopping and may form growths called tumor. This tumor starts spreading to different of body.

Tumors are of two types benign and malignant where benign (noncancerous) is the mass of cell which lack in ability to spread to other part of the body and malignant (cancerous) is the growth of cell which has ability to spread in other part 3 of body this spreading of infection is called metastasis There is various type of cancer like Lung cancer, leukemia, and colon cancer etc.

The incidence of lung cancer has significantly increased from the early 19th century. There is various cause of lung cancer like smoking, exposure to radon gas, secondhand smoking, and exposure to asbestos etc.

## 1.9 OBJECTIVE

1. Input Design is the process of converting a user-oriented description of the input into a computer-based system. This design is important to avoid errors in the data input process and show the correct direction to the management for getting correct information from the computerized system.

2. It is achieved by creating user-friendly screens for the data entry to handle large volume of data. The goal of designing input is to make data entry easier and to be free from errors. The data entry screen is designed in such a way that all the data manipulates can be performed. It also provides record viewing facilities.

3. When the data is entered it will check for its validity. Data can be entered with the help of screens. Appropriate messages are provided as when needed so that the user will not be in maize of instant. Thus the objective of input design is to create an input layout that is easy to follow.

# CHAPTER 2

# LITERATURE REVIEW

## 2.1 PROLOGUE

Due to its high death rates, cancer is still regarded as a serious disease in the twenty-first century. When it comes to morbidity and mortality, lung cancer is the most common type of cancer [2000, 2018]. Lung cancer can be divided into two primary types: small cell lung cancer (SCLC) and non-small-cell lung cancer (NSCLC). A subcategorization into lung squamous cell carcinoma (LUSC) and lung adenocarcinoma (LUAD) is also utilized for non-small-cell lung cancer. Roughly 85% of lung cancer cases are of these types of cancer [2022]. More detailed classification of lung cancers, such as LUSC, LUAD, and SCLC, is crucial for the prognosis of lung cancer when compared to the diagnosis of benign and malignant.

The effectiveness of treatment and, consequently, the survival rate of patients are directly impacted by accurately classifying the type of lung cancer at the time of early diagnosis [2000, 2015]. Both computed tomography (CT) and Positron Emission Tomography (PET) are popular noninvasive diagnostic imaging methods for lung cancer diagnosis specifically, as well as for clinical diagnosis generally [2015]. The gold standard for classifying lung cancer is immunohistochemical analysis. The invasive tissue biopsy required for this procedure, however, carries the risk of a delayed diagnosis and consequently worsening of the patient's pain.

Numerous studies on the automatic diagnosis of lung cancer were made possible by developments in artificial intelligence research. Three broad categories can be used to classify the use of data in lung cancer type classification: pathological images and CT and PET image data [2020]. The renowned data science community Kaggle gives users access to high-quality CT scans and asks them to differentiate between benign and malignant pulmonary nodules.

For these tasks, Kaggle competitions consistently yield top-notch deep learning strategies [2021, 2022]. Studies are no longer restricted to classifying benign and malignant nodules, and data sets are no longer restricted to CT images, thanks to advancements in the field of automatic lung cancer diagnosis [2019]. Wu et al. [2017] make use of quantitative imaging features.

Quantitative features based on PET images can effectively characterize intratumor heterogeneity and complexity, as demonstrated by the use of statistical, histogram-related, morphological, and textural features from PET images to predict the distance metastasis of NSCLC.

Two recent studies suggest using deep learning on pathological images to categorize transcriptome subtypes of LUAD [2020] and NSCLC and SCLC [2018]. The variety of imaging modalities used in the diagnosis of lung cancer further contributes to the intricacy of the clinical diagnosis [2018,2021]. Prior studies have demonstrated that deep learning techniques can combine various features in addition to using feature distribution patterns from various pulmonary imaging modalities. to accomplish the computer-assisted diagnosis. While some methods use a parallel CNN architecture to extract multiple features of various imaging modalities, Liang et al. [2018] use multichannel techniques to predict the IDH genotype from PET/CT data using a CNN [2015, 2022].

The different feature distributions and possible pathological features must be taken into account, the classification of the three types of lung cancer from medical images is more suited to represent a fine-grained image recognition problem than the classification of benign and malignant. The deep learning framework is vulnerable to feature noise since the lesion region is a tiny portion of the entire image, while fine-grained features must be extracted from the image. In fine-grained problems, the majority of approaches based on different deep learning frameworks have shown to have a certain bottleneck.

The attention mechanism from the two dimensions (channel and spatial) of the feature representation is primarily used in the prior research to address this issue. While the spatial attention mechanism makes sure that noise is reduced by weighting feature representation spatially [2021–2018], the channel attention mechanism simulates the relationship between feature channels [2020]. The spatial attention mechanism has been applied to medical images thus far. processing to improve features that have been extracted [2022, 2018].

Pulmonary disease has been identified and categorized using the channel attention mechanism [2021, 2020]. These attention mechanisms are presented in a way that shows the origin of distinctive noise from various angles. The deep learning model based on the multimodality dataset still struggles with fine-grained problems because there aren't many relevant studies on how to apply the attention mechanism more successfully on images with different imaging modalities.

Numerous studies have already been proposed for the prediction of cancer by different researchers. For example, Palani et al. [2020] proposed IoT-based predictive modeling using fuzzy C mean clustering for segmentation and an incremental classification algorithm that uses decision trees and association rule mining for classification to classify the tumor sets. Based on the output of the incremental classification model, a convolutional neural network has been applied with additional features to predict whether the tumors are benign or malignant.

Lynch et al., [2021] Various machine learning algorithm are implemented for predicting the survivability rate of person, performance is measured based on root mean square error. Each model is trained using 10-fold cross validation, as the parameters are preprocessed by assigning default value so cross 6 validation is used for avoiding over fitting.

FENWA et al., [2022] proposed a model whether feature like contrast, brightness from the image dataset is extracted using texture based feature extraction and on that two type of machine learning algorithm are applied one is artificial neural network another one is support vector machine and then performance has been evaluated on both the algorithm to compare which algorithm is giving more accuracy.

Öztürk et al., [2022] proposed a model where a five type of feature extraction techniques were used in individual classification algorithm to predict at which features extraction technique which machine learning algorithm is giving more accuracy.

Jin et al., [2019] proposed a model where the original image is first converted into binary image the erosion and dilution has been operated on that image after that image has been segmented on the segmented image region of interest extraction is applied to identify volume or size of the tumor and after extraction convolutional neural network is applied with softmax classification layer to recognize the tumor is cancerous or not.

Sumathipala et al., [2017] proposed a model where the image data are taken from LIDC-IDRI, after collecting the image data image filtration has been implemented, filtration is done based on the patient who went through biopsy and module level is equal to 30 and then images whose module level is equal to 30 is segmented and then Logistic regression and random forest has been applied for prediction.

## 2.2 METHODOLOGY

The application of DL techniques to the five main cancer types—liver, brain, lung, breast, and cervical is where this review begins.

**Lung Cancer**

In India, lung cancer is the most common and aggressive type with the highest death rates. Only 8% of patients with non-small cell lung cancer (NSCLC) have a 5-year survival rate [2020]. 18.4% of all deaths and 11.6 percent of all lung cancer cases from lung cancer are documented in the GLOBOCON report for 2020. More people have lung cancer than any other type of cancer, including those of the breast, liver, cervical, skin, etc.

If lung cancer is detected early, patients' chances of survival can be improved. Since most of the time, late-stage cancer detection precludes surgical treatment, nonsurgical methods such as immunotherapy, radiotherapy, and chemotherapy. In the classification and early detection of lung nodules using CT scans, computer vision techniques produced significant results [2020].

### 2.2.1 Automatic lung cancer detection using Medical images

Acquiring and preprocessing medical images, typically CT scans, is the first step in automatic detection systems. Digital image processing, which extracts useful features like tumor size, shape, and location, and machine learning, which allows computers to learn patterns from massive labeled datasets, provide the theoretical underpinnings. Commonly employed methods include the Random Forest Classifier and deep learning frameworks, which include customized Convolutional Neural Networks (CNNs). These models handle both classification (benign vs. malignant) and localization (determining the tumor's position within the scan) and are trained to identify patterns suggestive of cancer.

Window setting optimization, slice thickness modification, and data augmentation are crucial pre-processing steps that help guarantee models can generalize across various patient populations and imaging protocols. Shape, texture, and other visual information are converted into data that algorithms can process through feature extraction. The accuracy and efficiency of detection are further increased by modern models' incorporation of attention mechanisms, robust loss functions, and anchor-free detection for tiny tumors.

The performance of these systems is objectively assessed using evaluation metrics such as accuracy, recall, precision, and mean average precision (mAP). Modern models have demonstrated outstanding performance in clinical trials, with accuracy rates exceeding 93%, precision reaching 98%, and consistent recall for both cancerous and non-cancerous cases.

Professionals can more easily identify lung cancer in its early stages, when treatment is most effective, thanks to these methods' significant speed and consistency improvements over manual review. Recent studies have shown that models such as YOLOv11 combine localization, classification, and detection into a single system, outperforming and outperforming older algorithms. Tumor identification and volume estimation are further improved by ensemble models and segmentation networks like U-Net, which are essential for tracking progression and developing treatment plans.

Grad-CAM visualizations and other Explainable AI (XAI) features are being incorporated more frequently to increase transparency. These foster trust and facilitate improved patient communication by letting clinicians see which image regions influenced the model's decision. Despite significant progress, obstacles still exist. Handling class imbalances (datasets with fewer non-cancer cases than cancer cases), guaranteeing generalizability to unobserved patient populations, and maximizing computational resources for real-time clinical integration are among the challenges. By using creative architectures, well-rounded training methods, and ongoing annotated dataset expansion, current research seeks to address these issues.

### 2.2.2 Deep Learning for Lung Cancer Detection

The lung cancer has the highest rates of morbidity and death of any cancer type, early detection is essential to saving lives. In order to visualize lung tumors and analyze characteristics such as size, location, and shape, computed tomography (CT) scans are essential. Recently, the field of image-based diagnosis has undergone a transformation thanks to deep learning, a branch of artificial intelligence that uses layered neural networks to simulate the human brain. These models are excellent at identifying patterns and can reveal intricate connections in imaging data that conventional techniques might miss.

CT scan features, such as tumor border, texture, sharpness, and density, are first extracted and analyzed by traditional machine learning methods like the Random Forest classifier. In order to increase accuracy and decrease overfitting, ensemble algorithms like Random Forest combine predictions from several decision trees. On the other hand, deep learning models—particularly Convolutional Neural Networks (CNNs)—have demonstrated superiority for large, diverse CT image datasets due to their ability to automatically identify and learn hierarchical image features with little assistance from humans, resulting in more reliable and scalable cancer detection systems.

The main clinical benefit of deep learning for lung cancer detection is consistent, quick evaluation, which lowers human error and helps radiologists identify tumors at earlier, more manageable stages. Additionally, the technique helps with planning interventions, tracking patient response over time, and quantifying tumor progression. Automated systems frequently outperform conventional rule-based diagnostic techniques in detecting both overt and covert indications of lung cancer.

Class imbalance (more positive than negative cases), generalizability to other populations, and the requirement for sizable, varied annotated datasets to train trustworthy models are still major obstacles. Deep learning techniques may have trouble with novel imaging protocols or uncommon subtypes of lung cancer because they rely on high-quality labeled data. Data augmentation techniques, transfer learning (retraining models using fresh data), and ongoing model parameter updates are how researchers are tackling these issues.

Going forward, there is a lot of promise in incorporating deep learning into standard screening and diagnostic processes. Lung cancer detection is becoming more accessible, accurate, and efficient for better patient outcomes thanks to ongoing improvements in explainability, computational efficiency, and open collaboration between data scientists and clinicians.

### 2.2.3 Review On Feature Extraction Methods

Feature extraction is one of the most important processes in medical image analysis and computer-aided diagnosis, especially in the field of lung cancer prediction. Medical images such as CT scans or X-rays contain large amounts of raw data that cannot be directly used for classification by machine learning algorithms. These images are high-dimensional, noisy, and often contain redundant information. The process of feature extraction converts this raw data into a smaller set of meaningful descriptors that represent the most important aspects of a lung nodule, such as its size, texture, shape, and intensity distribution. This step reduces computational complexity, eliminates irrelevant information, and provides discriminative power for classifiers to distinguish between benign and malignant cases.

Different approaches have been proposed over the years for extracting features in lung cancer studies. Texture-based methods are widely used since they capture surface irregularities and fine details of lung nodules that may not be visible to the naked eye. Cancerous tissues usually appear heterogeneous and irregular, while non-cancerous tissues have smoother and more uniform textures. Statistical measures derived from the relationship of pixel intensities, such as contrast, correlation, energy, and homogeneity, provide valuable insights into tumor tissue characterization. Shape-based methods, on the other hand, analyze

the geometric properties of lung nodules. Tumors that are malignant tend to show irregular borders, lobulation, or spiculation, while benign nodules are typically circular and smooth. By calculating properties such as perimeter, area, compactness, or irregularity, researchers can differentiate abnormal growth patterns more effectively.

Another important category is intensity-based features, which are derived from the distribution of pixel values in lung images. By analyzing brightness levels, pixel density, and statistical variation in intensity, one can estimate the internal structure of nodules. For instance, the measurement of Hounsfield Units in CT images provides details on tissue density, which can be a strong indicator of malignancy. Transform-based methods add another dimension to feature extraction. By representing images in alternative mathematical forms such as wavelet or Fourier transforms, they are able to capture frequency-domain information that is not visible in the spatial domain.

These methods are particularly useful for dimensionality reduction, noise elimination, and the retention of essential diagnostic features.With the rise of deep learning, feature extraction has undergone a major transformation. Instead of manually designing descriptors, deep learning techniques such as convolutional neural networks are able to automatically learn hierarchical features directly from the image data. At lower layers, simple patterns like edges or corners are detected, while deeper layers capture more complex patterns related to tumor shape and internal structure.

Transfer learning using pre-trained networks like VGGNet, ResNet, or Inception has also become popular, as it allows researchers to apply knowledge learned from large general datasets to medical imaging tasks, thereby improving performance even with limited training data. Autoencoders are another deep learning approach that can learn compressed representations of images without supervision, making them useful for dimensionality reduction and unsupervised feature learning.

Recent studies suggest that the best results are achieved when different types of features are combined rather than relying on a single method. Hybrid feature extraction strategies that integrate handcrafted features such as texture, shape, and intensity with automatically learned deep features are proving highly effective in lung cancer diagnosis. This combination ensures that models benefit from both interpretability and high predictive accuracy. Traditional handcrafted features are easier to understand and explain, while deep features capture complex variations that may not be visible to conventional statistical methods. By merging these approaches, researchers are able to design robust systems that can perform accurately even in the presence of noisy or unbalanced datasets.

A comparative review of existing methods shows that traditional features are simple to compute and effective in small datasets, but they may not capture subtle variations in cancer tissues. Transform-based methods improve feature quality but sometimes lose interpretability. Deep learning-based features are highly accurate but computationally expensive and often require large labeled datasets. Hybrid features provide the most balanced outcome, combining accuracy, robustness, and interpretability. Therefore, in modern lung cancer research, hybrid feature extraction has become the preferred strategy for building reliable computer-aided diagnostic systems.

# CHAPTER 3

# SYSTEM REQUIREMENT AND SPECIFICATION

## 3.1 HARDWARE SEPECIFICATION

The hardware requirements may serve as the basis for a contract for the implementation of the system and should therefore be a complete and consistent specification of the whole system. They are used by software engineers as the starting point for the system design. It shows what the systems do and not how it should be implemented.

- PROCESSOR: Intel(R) CORE(TM) i3-2328M CPU
- RAM: 4.00GB
- HARD DISK: 465.3GB

## 3.2 SOFTWARE SEPECIFICATION

The software requirements are the specification of the system. It is a set of what the system should do rather than how it should do it. The software requirements provide a basis for creating the software requirements specification. It is useful in estimating cost, planning team activities, performing tasks and tracking the team's and tracking the team's progress throughout the development activity.

- OPERATING SYSTEM: Microsoft Windows 8.1 Pro
- SYSTEM TYPE: X86-based PC, intel CORE i3
- SYSTEM MODEL: HP 2000 Notebook PC

## 3.3 SOFTWARE DESCRIPTION

The proposed lung cancer prediction system is implemented using machine learning techniques that process medical data and classify patients based on their likelihood of having lung cancer. The software is designed to assist medical practitioners by providing accurate predictions from clinical datasets and CT scan features.

The system consists of three main modules:data preprocessing, feature extraction, and classification. In the preprocessing stage, the dataset is cleaned by handling missing values, removing inconsistencies, and normalizing the data so that all attributes fall within a uniform range. This step ensures that the input data is reliable and suitable for further processing.

In the feature extraction stage, important attributes such as patient age, smoking history, coughing, chest pain, and shortness of breath, as well as image-based features from CT scans, are derived. These features represent the key factors that influence lung cancer detection. By reducing irrelevant data and focusing on the most informative features, the system improves efficiency and accuracy.

The classification stage applies machine learning algorithms such as Random Forest Classifier and Histogram Gradient Boosting Classifier to the extracted features. These algorithms are chosen because they handle both categorical and numerical data effectively, reduce overfitting, and provide robust predictions. The classifiers analyze the features and predict whether the patient is likely to have lung cancer (YES) or not (NO). The performance of the system is evaluated using metrics such as accuracy, precision, recall, and F1-score.

The software is implemented in Python, using libraries such as Scikit-learn, Pandas, NumPy, and Matplotlib for model training, data handling, and visualization. The user interface is designed in a simple, interactive format, allowing medical users to input patient data and receive prediction results.

This software system helps in reducing diagnostic errors, speeds up the process of early cancer detection, and provides decision support to healthcare professionals. By integrating machine learning algorithms with medical datasets, the software ensures a more reliable and efficient approach to predicting lung cancer at an early stage.

# CHAPTER 4

# SYSTEM ANALYSIS

## 4.1 EXISTING SYSTEMS

Existing systems for lung cancer prediction and diagnosis combine traditional clinical practice with a variety of computer-aided methods. At the clinical level, radiologists and oncologists rely on imaging modalities such as chest X-ray and computed tomography (CT) scans, together with patient history (smoking, age, symptoms), to make diagnostic decisions. This human-led workflow remains the gold standard because of clinicians' ability to integrate imaging, symptoms, and patient context, but it is time-consuming and subject to inter-observer variability — different experts can disagree on whether a nodule looks suspicious. To reduce missed cases and speed up diagnosis, a number of computerized systems have been developed and deployed.

Early computerized approaches are based on classical image processing and machine learning. These systems typically begin with preprocessing (noise reduction, lung segmentation) followed by handcrafted feature extraction: texture descriptors (e.g., GLCM statistics), intensity/histogram measures, and geometric/shape descriptors (area, perimeter, irregularity). Extracted features are then fed to conventional classifiers such as logistic regression, k-nearest neighbors, support vector machines, decision trees, or Random Forests. Such pipelines are relatively interpretable and inexpensive to run, and they perform well on small to moderate datasets. However, their performance depends heavily on the quality of the handcrafted features and on manual tuning of preprocessing steps.

As computational power and dataset sizes increased, ensemble methods and gradient boosting machines (for example, XGBoost or histogram-based gradient boosting) became common because they can handle heterogeneous feature sets and imbalanced data more robustly. These systems often outperform single classical classifiers and are used when clinical and extracted image features are combined (multimodal data). Many research prototypes and some clinical decision-support tools use ensembles to provide probability scores (risk estimates) rather than binary outputs, which helps clinicians prioritize cases for further review.

The most transformative change in recent years comes from deep learning–based systems. Convolutional neural networks (CNNs) are now widely used to analyze chest CTs and X-rays because they learn hierarchical feature representations directly from images, eliminating the need for manual feature engineering. End-to-end CNN models can perform nodule detection,

segmentation, and malignancy classification within a single framework. Transfer learning (fine-tuning networks pretrained on large natural image datasets) is commonly used to improve performance when medical training data are limited. Deep models tend to achieve higher accuracy than classical methods on large annotated datasets and have been integrated into experimental clinical workflows and research platforms.

Hybrid systems combine the strengths of both approaches by fusing handcrafted features and deep features, or by combining image-derived features with clinical data (age, smoking history, symptoms, lab results). These multimodal systems produce more robust predictions because they allow the model to use both interpretable clinical indicators and the nuanced image patterns that deep networks capture. Many recent studies show that hybrid pipelines — for example, a CNN for image feature extraction plus a gradient boosting classifier that also uses clinical features — yield improved sensitivity and specificity.

Commercial and research-grade computer-aided detection/diagnosis (CAD) systems exist that assist radiologists by flagging suspicious nodules, measuring size and volume, and providing malignancy risk scores. These tools are typically cleared for clinical use in certain regions and used as a second reader: they do not replace radiologists but help reduce oversight and double-check borderline cases. The practical deployment of CAD systems highlights important non-technical factors such as integration with picture-archiving and communication systems (PACS), user interface design, and explainability of outputs.

Despite progress, existing systems face persistent limitations. Many models show performance drops when applied to images from different scanners, hospitals, or patient populations (domain shift). Limited labeled medical data and class imbalance (fewer benign versus malignant examples or vice versa) make training robust models difficult. Interpretability is another concern: clinicians need understandable explanations for model predictions to trust them in high-stakes decisions. Finally, regulatory, privacy, and workflow-integration issues affect real-world adoption even when model accuracy is high.

### 4.2 PROPOSED SYSTEM

A methodical machine learning workflow for lung cancer prediction begins by uploading and carefully preparing the cancer dataset. During data pre-processing, the dataset is cleaned by removing missing values, inconsistencies, and irrelevant information to ensure that the input is reliable and meaningful for modeling. Applying a normalization technique further ensures that all features are on a consistent scale, which is essential for accurate and efficient analysis by machine learning models.

After normalization, the data is split into training and testing sets, which separates samples used to build the models from those used to evaluate them. This division helps prevent overfitting and provides a realistic measure of model performance on new, unseen data. The Random Forest Classifier (RFC) is employed as the main predictive algorithm in this workflow due to its strength in managing complex datasets, such as those found in medical research. RFC works by creating an ensemble of decision trees and aggregating their outputs, making it robust and less prone to overfitting.

To enhance the modeling and allow performance comparison, two additional classifiers are incorporated: Support Vector Classifier (SVC) and Histogram Gradient Boosting Classifier (HGBC). SVC seeks the best boundary between classes—cancer and non-cancer—by analyzing data in high-dimensional space, which is useful in situations with intricate relationships among features. HGBC builds models sequentially, focusing on correcting errors from previous trees, which improves predictive accuracy, particularly when complex patterns are present.

Once the models are constructed, predictions are generated and evaluated using a range of performance metrics. The accuracy score provides a quick measure of overall correct classification, but a confusion matrix is also constructed to offer detailed insight into true positives, true negatives, false positives, and false negatives. This matrix is crucial for identifying where the model excels and where it might need improvement, especially in medical diagnostics where false predictions carry significant consequences.

Advanced metrics like precision, recall, and the F1-score curve are calculated to assess the model's reliability further. Precision focuses on the proportion of positive identifications that are actually correct, while recall measures the proportion of true positives detected among all actual cases of cancer. The F1-score harmonizes these two metrics, balancing the need to identify cancer cases accurately while minimizing incorrect diagnoses. These metrics are particularly valuable in healthcare, where both sensitivity to disease and avoidance of unnecessary treatment are critical.

**I. Random forest Classifier(RFC)**

The Random Forest machine learning algorithm effectively reduces overfitting and increases accuracy by classifying lung cancer risk using multiple decision trees. To construct each tree, the model selects samples and features at random, and the majority vote from all trees determines the final output. It is perfect for medical datasets because it can manage both numerical and categorical data in practice. According to studies, Random Forest predicts lung cancer early with high accuracy (up to 96.77%). Additionally, it draws attention to significant

risk factors like age and smoking, which enhances the interpretability of the model. Because of this, it is a reliable and strong tool for determining the presence of lung cancer in clinical settings.

**II. Histgradient Boosting Classifier (HGBC)**

The Histgradient Boosting Classifier is a sophisticated machine learning algorithm that predicts lung cancer by successively constructing decision trees, each of which fixes the mistakes of the one before it. Histogram binning, which arranges features into fixed intervals, increases training speed and memory usage. Large datasets, such as medical records, are a good fit for this approach. It can handle class imbalance to increase predictive accuracy and is very good at handling continuous and categorical data. Research indicates that combining methods such as SMOTE with boosting algorithms improves model performance even more. Clinicians can better understand risk factors by using the algorithm's interpretable results in addition to its high accuracy. Metrics like accuracy, precision, recall, F1-score, and ROC AUC are used to assess its performance, guaranteeing its dependability in the diagnosis of lung cancer.

**III. Support Vector Machine (SVM)**

A machine learning algorithm called Support Vector Machine (SVM) is used to predict lung cancer by determining the optimal boundary (hyperplane) that divides various classes, such as cancerous and non-cancerous cases. Clinical features and medical images are examples of high-dimensional data that it performs well with. To deal with non-linear relationships in the data, SVM can employ various kernel functions. According to studies, SVM models can classify lung cancer with high accuracy—some have been shown to achieve up to 97% accuracy. When properly trained, it produces dependable predictions and manages both small and large datasets effectively. The performance of SVM can be further enhanced by optimization strategies like hybrid models and parameter tuning. SVM is frequently used in cancer diagnosis research because of its interpretability and robustness.

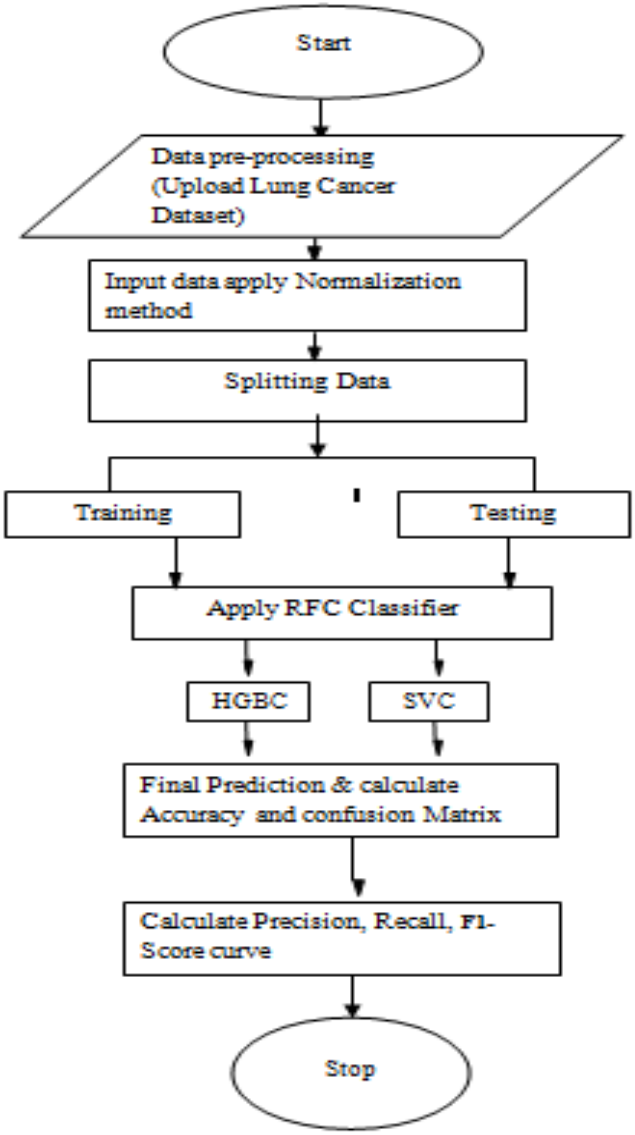


**Figure 5: Flow Diagram of Proposed work**

# CHAPTER 5

# SYSTEM DESIGN

## 5.1 ARCHITECTURE DIAGRAM

The following algorithm describes the step by step approach for the proposed model

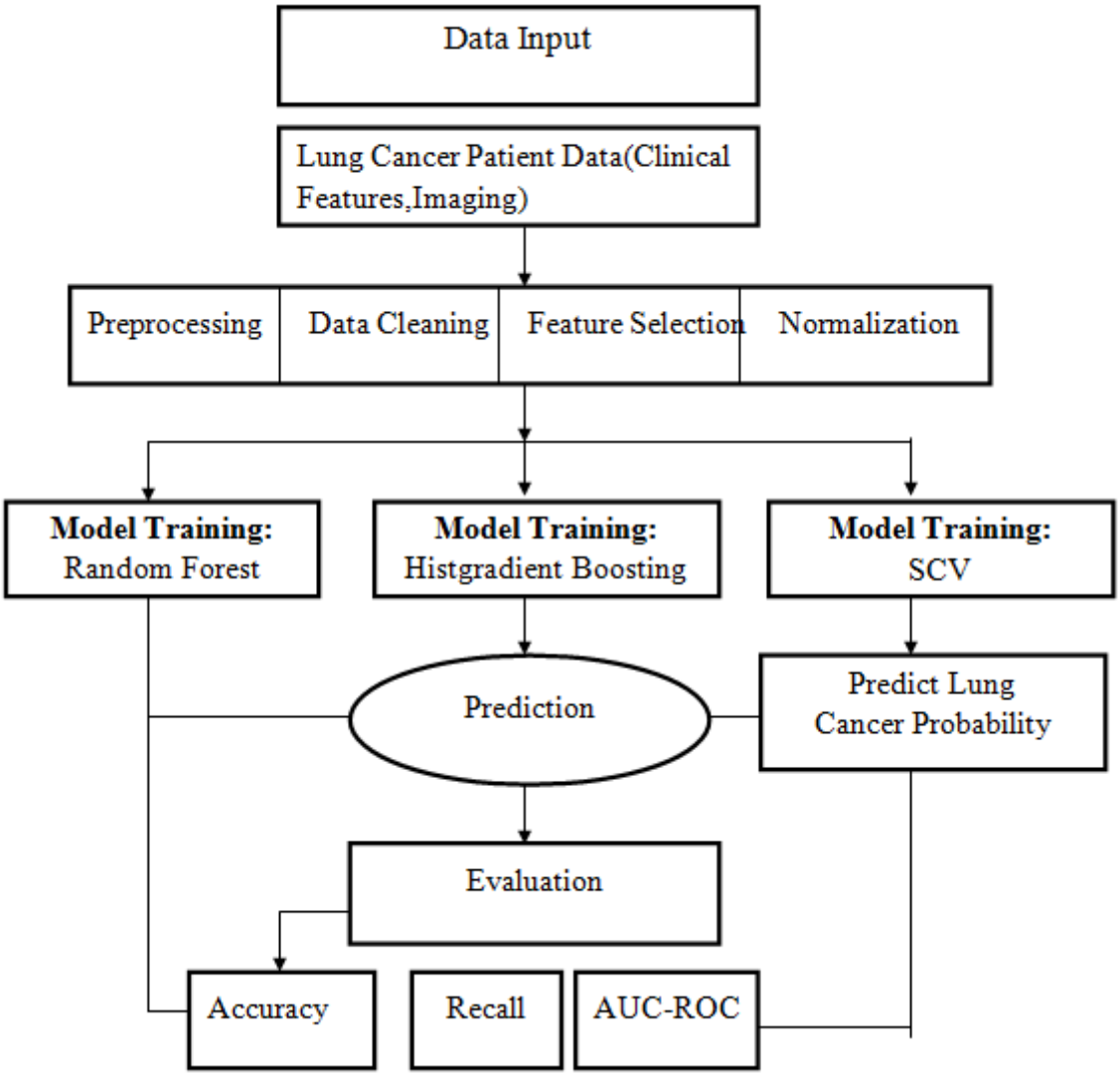


**Figure 6: Implementation Model**

1. Input: The process begins with collecting lung cancer patient data, including clinical features and imaging results.

2. Data Pre-processing:

- Data Cleaning: Removes missing, inconsistent, or noisy data.
- Feature Selection: Identifies the most relevant variables affecting lung cancer prediction.
- Normalization: Scales the data to ensure consistent ranges across features

3. Model Training: Three algorithms are trained separately on the processed data

- Random Forest Classifier: A tree-based ensemble model that improves prediction accuracy.
- Histogram Gradient Boosting (HGB): Builds models sequentially to correct previous errors and enhance accuracy.
- Support Vector Classifier (SVC): Finds the best hyperplane to separate cancer and non-cancer cases.

4. Prediction: Each trained model predicts the probability of lung cancer for new or test samples.

- Evaluation: Model performance is assessed using key metrics:
- Accuracy: Measures overall correctness of predictions.
- Recall: Indicates how well the model identifies actual cancer cases.
- AUC-ROC: Evaluates the model's ability to distinguish between positive and negative cases.

**PRE-PROCESSING**

In the analytics process of lung cancer prediction, pre-processing is one of the most important tasks. It is often observed that this stage consumes more than half of the total time in the development of a machine learning model.

The raw patient data collected from medical records may not always be clean or in the required format. Issues such as missing values, noisy attributes, and duplicate entries can affect the reliability of predictions. If used directly, such data may lead to inaccurate or misleading results.

Therefore, pre-processing is applied to convert patient details into a suitable format for machine learning models. This stage addresses inconsistencies in the dataset by handling missing values, removing duplicates, and reducing noise. It also includes attribute

scaling(e.g., normalizing age, tumor size), dataset splitting (into training and testing sets), and data encoding (converting categorical values like smoker/non-smoker into numerical form).

By performing these steps, the dataset becomes structured, consistent, and ready for classification algorithms such as Random Forest and HistGradient Boosting Classifier. Effective pre-processing significantly improves the accuracy, precision, and recall of the lung cancer prediction system, ensuring reliable early diagnosis.

## 5.2 DATASET DESCRIPTION

**Breathlessness In Lung Cancer**

Dyspnoea, or shortness of breath, is one of the most frequent and distressing symptoms in lung cancer and significantly impairs patients' quality of life at virtually any stage of the disease. Over 70 percent of people with advanced lung cancer report experiencing dyspnoea at some point, and it may occur during early tumor growth or later, advanced stages. It is particularly prevalent in the terminal phase, with up to 80 percent of patients affected in their final weeks of life. Accurate assessment is complicated because dyspnoea is profoundly subjective, meaning it may not correlate with clinical observations (such as normal lab tests or oxygen levels). Patient self-report is considered the gold standard for monitoring severity and impact, with validated questionnaires and patient-reported outcome measures used in routine care.

**Tumour Obstruction:**

Tumor obstruction in the lungs can cause significant complications beyond airflow blockage. It raises the risk of respiratory infections such as pneumonia by trapping mucus and bacteria behind the obstruction. These infections can become difficult to treat and may lead to recurrent or chronic lung infections. The immune system may also be weakened by cancer and its treatments, making infections more severe. Structural changes caused by tumors prevent normal airway clearance, worsening mucus buildup. Other complications include lung abscess and accumulation of pus in the pleural cavity (empyema). Managing these infections often requires not only antibiotics but also interventions to relieve airway obstruction and promote drainage. Early detection and treatment are critical to prevent deterioration in lung function and improve outcomes for patients.

**Pleural Effusion:**

Pleural effusion can cause a persistent, often dry cough due to irritation of the lung and pleural surfaces. The buildup of fluid compresses the lung, making it unable to fully expand during breaths, leading to shortness of breath and chest tightness. Large or rapidly increasing

effusions may cause fever or chills if the fluid becomes infected. Patients may experience difficulty breathing when lying flat, known as orthopnea, which disrupts sleep and daily comfort. This condition also increases the effort needed to breathe, causing fatigue and anxiety. The fluid accumulation may compress nearby blood vessels, reducing oxygen delivery to the body and worsening breathlessness. Effusions can significantly limit physical activity and quality of life, requiring medical intervention to drain the fluid and relieve symptoms.

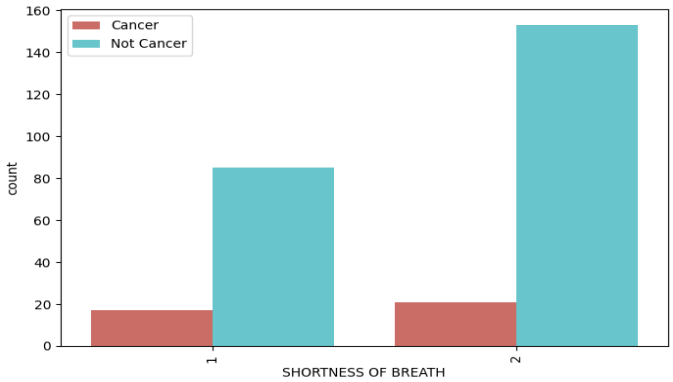


**Figure 7: shortness of breath**

**Smoking Distribution**

smoking is a significant factor in studies that forecast the occurrence of lung cancer because it raises the risk of the disease.

**Table1: Smoking Distribution**

| SOMKING | DESCRIPTION | PERCENTAGE | INTERPRETATION |
|---|---|---|---|
| 2 | Smoker | 54.30% | Majority are smokers |
| 1 | Non-Smoker | 45.70% | Slightly fewer non-smokers |

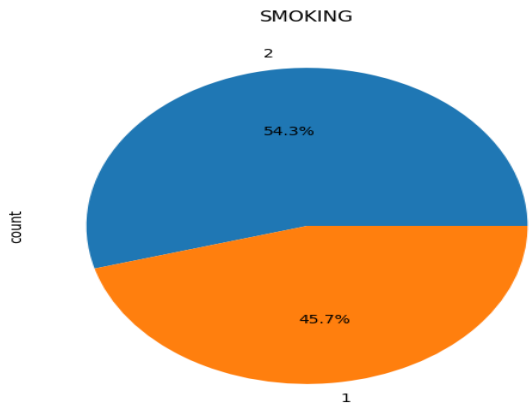


**Figure 8: Smoking**

The data is separated into two groups: non-smokers (code 1) and smokers (code 2). The percentage distribution facilitates comprehension of the smoking prevalence among the participants.According to the data, 54.30% of people smoke, while 45.70% do not. Given that most of the population under study smokes, this indicates a high prevalence of smoking behavior in the dataset. The interpretation of this distribution is significant because it confirms medical evidence that smoking is a significant risk factor for lung cancer.The slightly lower percentage of people who don't smoke implies that even

**Table2: Dataset Research**

| | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | GENDER | AGE | SMOKING | YELLOW_F | ANXIETY | PEER_PRE | CHRONIC | FATIGUE | ALLERGY | WHEEZIN( | ALCOHOL | COUGHIN( | SHORTNE: | SWALLOW | CHEST PAI | LUNG_CANCER |
| 2 | M | 69 | 1 | 2 | 2 | 1 | 1 | 2 | 1 | 2 | 2 | 2 | 2 | 2 | 2 | YES |
| 3 | M | 74 | 2 | 1 | 1 | 1 | 2 | 2 | 2 | 1 | 1 | 1 | 2 | 2 | 2 | YES |
| 4 | F | 59 | 1 | 1 | 1 | 2 | 1 | 2 | 1 | 2 | 1 | 2 | 2 | 1 | 2 | NO |
| 5 | M | 63 | 2 | 2 | 2 | 1 | 1 | 1 | 1 | 1 | 2 | 1 | 1 | 2 | 2 | NO |
| 6 | F | 63 | 1 | 2 | 1 | 1 | 1 | 1 | 1 | 2 | 1 | 2 | 2 | 1 | 1 | NO |
| 7 | F | 75 | 1 | 2 | 1 | 1 | 2 | 2 | 2 | 2 | 1 | 2 | 2 | 1 | 1 | YES |
| 8 | M | 52 | 2 | 1 | 1 | 1 | 1 | 2 | 1 | 2 | 2 | 2 | 2 | 1 | 2 | YES |
| 9 | F | 51 | 2 | 2 | 2 | 2 | 1 | 2 | 2 | 1 | 1 | 1 | 2 | 2 | 1 | YES |
| 10 | F | 68 | 2 | 1 | 2 | 1 | 1 | 2 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | NO |
| 11 | M | 53 | 2 | 2 | 2 | 2 | 2 | 1 | 2 | 1 | 2 | 1 | 1 | 2 | 2 | YES |
| 12 | F | 61 | 2 | 2 | 2 | 2 | 2 | 2 | 1 | 2 | 1 | 2 | 2 | 2 | 1 | YES |
| 13 | M | 72 | 1 | 1 | 1 | 1 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 1 | 2 | YES |
| 14 | F | 60 | 2 | 1 | 1 | 1 | 1 | 2 | 1 | 1 | 1 | 1 | 2 | 1 | 1 | NO |
| 15 | M | 58 | 2 | 1 | 1 | 1 | 1 | 2 | 2 | 2 | 2 | 2 | 2 | 1 | 2 | YES |
| 16 | M | 69 | 2 | 1 | 1 | 1 | 1 | 1 | 2 | 2 | 2 | 2 | 1 | 1 | 2 | NO |
| 17 | F | 48 | 1 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 1 | 2 | 2 | 2 | 1 | YES |
| 18 | M | 75 | 2 | 1 | 1 | 1 | 2 | 1 | 2 | 2 | 2 | 2 | 2 | 1 | 2 | YES |
| 19 | M | 57 | 2 | 2 | 2 | 2 | 2 | 1 | 1 | 1 | 2 | 1 | 1 | 2 | 2 | YES |
| 20 | F | 68 | 2 | 2 | 2 | 2 | 2 | 2 | 1 | 1 | 1 | 2 | 2 | 1 | 1 | YES |
| 21 | F | 61 | 1 | 1 | 1 | 1 | 2 | 2 | 1 | 1 | 1 | 1 | 2 | 1 | 1 | NO |
| 22 | F | 44 | 2 | 2 | 2 | 2 | 2 | 2 | 1 | 1 | 1 | 1 | 2 | 2 | 1 | YES |
| 23 | F | 64 | 1 | 2 | 2 | 2 | 1 | 1 | 2 | 2 | 1 | 2 | 1 | 2 | 1 | YES |
| 24 | F | 21 | 2 | 1 | 1 | 1 | 2 | 2 | 2 | 1 | 1 | 1 | 2 | 1 | 1 | NO |
| 25 | M | 60 | 2 | 1 | 1 | 1 | 1 | 2 | 2 | 2 | 2 | 2 | 2 | 1 | 2 | YES |

The purpose of this dataset is to examine the variables that affect lung cancer, with an emphasis on symptoms, personal information, and medical history. It is a valuable tool in medical research, especially when it comes to statistical or machine learning-based predictive modeling.A unique patient record is represented by each instance:

This attribute, which is coded as "M" for male and "F" for female, indicates the patient's gender. Because males and females differ in hormonal, behavioral, and genetic factors that affect cancer susceptibility, gender differences can affect the prevalence of lung cancer. The age of the patient is recorded numerically. Age is important because the risk of lung cancer usually rises with age because of the body's weakened ability to repair cells over time

and extended exposure to carcinogens.Here, smoking status is encoded numerically, most likely to indicate frequency or quantity (e.g., 1 = light smoker, 2 = heavy smoker). The biggest risk factor for lung cancer is smoking, which causes cellular changes in the lungs. The acronym most likely refers to yellow fingers, a condition linked to heavy smoking and tar buildup from cigarette smoke. One obvious indicator of tobacco use is yellowing of the fingers.

This field may be used to indicate whether anxiety symptoms are present or how severe they are (coded numerically). Through immunological suppression or lifestyle choices, psychological stress and anxiety may have an indirect impact on cancer risk or prognosis. refers to peer pressure, which may have an impact on actions like starting or continuing to smoke.

This variable might indicate how social factors influence health-related behavior. Lung diseases, including cancer, are at risk for chronic illnesses or conditions like COPD or chronic bronchitis. The presence or severity of this attribute is indicated by a numerical code. Lung cancer is among the many cancers that frequently exhibit fatigue, which is frequently associated with systemic effects or decreased oxygenation. Allergies may be noted, which could affect immunological responses and lung inflammation. A common respiratory symptom of lung diseases, wheezing may indicate airway blockage brought on by tumors.

# CHAPTER 6

# RESULT AND DISCUSSIONS

## 6.1 PRECISION TESTING

The performance of a lung cancer prediction model can be evaluated using important classification metrics such as accuracy, precision, recall, and F1-score. These metrics help assess the model's ability to identify both lung cancer and non-cancer cases. According to the model's precision of 0.50 for the "NO" class, which stands for non-lung cancer cases, only 50% of the cases that were predicted as "NO" were in fact correct.

**Recall:**

The recall of 0.25 indicates that only 25% of actual non-lung cancer cases were correctly identified.

**F1-score:**

The F1-score of 0.33 for this class suggests a poor balance between recall and precision. The low recall value suggests that the model might struggle to identify non-cancer cases due to a class imbalance, given the support (actual number of cases).

**Table3: Precision from the F1-srcoe**

| CLASS | PRECISION | RECALL | F1-SCORE | SUPPORT | MEANING |
|---|---|---|---|---|---|
| NO | 0.5 | 0.25% | 0.33 | 8 | Non-lung cancer cases |
| YES | 0.9 | 0.96% | 0.93 | 54 | Lung cancer cases |
| Overally Accuracy | 0 | 0 | 0.87 | 62 | Model correctly predicts 87% of all cases |

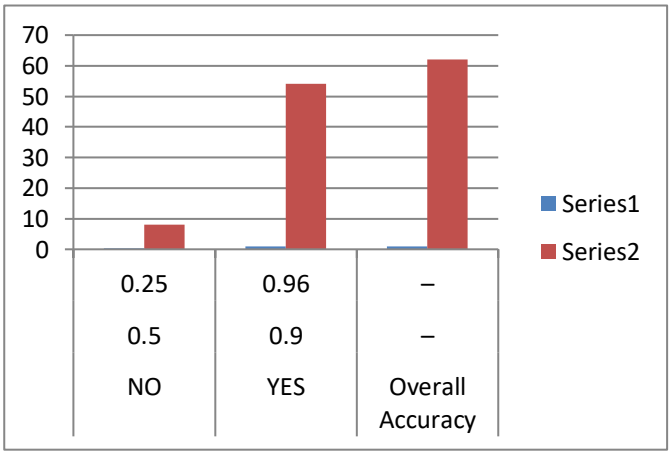


**Figure 9: Precision from the F1-srcoe**

## 6.2 ACCURACY TESTING

A machine learning model's performance on both training and test data is shown by the dataset accuracy results. One important evaluation metric is accuracy, which quantifies the percentage of correctly predicted cases among all samples. It displays the frequency of accurate predictions made by the model and is expressed as a percentage.

The model's training accuracy in this instance is 80.16%, which indicates that it correctly classified or predicted about 80 out of every 100 samples when tested on the data it was trained on. This shows that the model has done a respectable job of learning the relationships and patterns found in the training data. However, because a model can occasionally memorize the training data rather than generalizing from it—a problem known as overfitting—it does not always indicate how well the model will perform on unseen data.

With a test accuracy of 82.26%, the model performs marginally better when used with fresh, untested test data. This indicates that roughly 82 out of every 100 test samples are correctly predicted by the model. The model is said to generalize well and not overfit to the training dataset if the test accuracy is higher than the training accuracy. Instead of memorizing the key patterns, it might have learned them.

The model appears to achieve a good balance between bias and variance, as evidenced by the small difference (roughly 2%) between training and test accuracy. It does not overfit (too customized to training data) or underfit (too basic to capture relationships). A stable model that can produce accurate predictions on fresh data is demonstrated by this level of performance.

**Table 4: Model Accuracy on Training Vs Test Data**

| DATASET | ACCURACY(%) | INTERPRETATION |
|---|---|---|
| Training Data | 80.16% | The model correctly predicts~80 out of 100 training sample |
| Test Data | 82.26% | The model correctly predicts~80 out of 100 unseen sample |

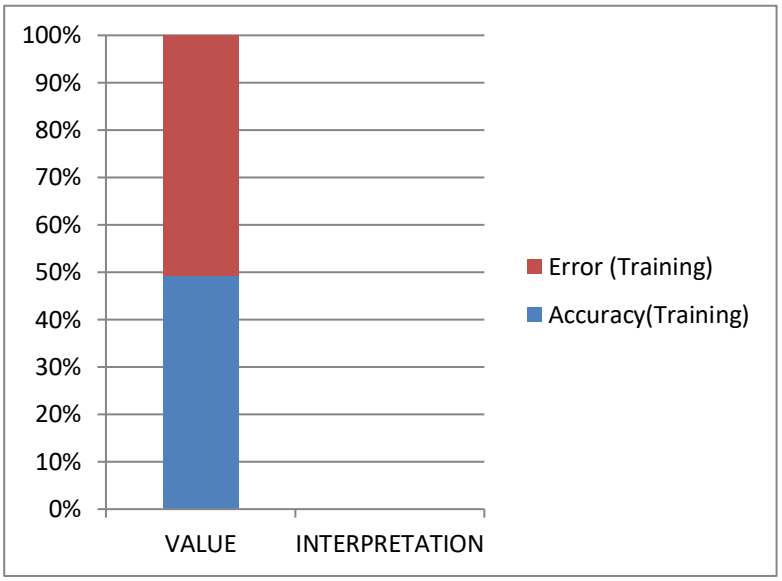


**Figure 10: Model Accuracy on Training Vs Test Data**

# CHAPTER 7

# TESTING METHODS

## 7.1 UNIT TESTING

While it is not uncommon for coding and unit testing to be carried out as two distinct stages, unit testing is usually carried out as part of the software life cycle's combined code and unit test process.

**Test methodology and strategy**

Functional tests will be meticulously written, and field testing will be done by hand.

**Verify your objectives**

- Every field entry must function properly.
- Pages from the discovered connection must be permitted.
- Pay attention to the entry screen, alerts, and responses.

**Features to look for:**

- Verify that submissions follow the correct format;
- Don't allow duplicate entries
- Every link will direct the user to the relevant page.

## 7.2 INTEGRATION TESTING

Software integration testing involves integrating two or more compatible software components gradually on a single platform in order to generate errors brought on by flaws in the device. The integration check's objective is to confirm whether parts or software Applications, such as parts of a software system or advanced software applications at the corporate level, communicate without errors.

**Test Results:** Every one of the aforementioned test cases was successfully completed. No flaws were discovered.

## 7.3 ACCEPTANCE TESTING

Acceptance by Users Any project's testing phase is crucial and necessitates significant end user participation. This guarantees that the framework satisfies the practical requirements as well.

**Test Results:** Every one of the aforementioned test cases was successfully completed. No flaws were discovered.

**Functional Test**

Functional tests offer methodical proof that the functions being tested are available in accordance with the technical and business requirements, system documentation, and user manuals.

The following are the main focus of functional testing:

- Valid Input: The recognized categories of valid input need to be approved.
- Invalid Input: Classes of input that have been identified as invalid must be rejected.
- Functions: The functions that have been identified must be used.
- Output: It is necessary to exercise the designated classes of application outputs.
- Systems/Procedures: It is necessary to invoke interacting systems or procedures.

Functional tests are organized and prepared with requirements, important functions, or unique test cases in mind. Furthermore, data fields, predefined processes, and subsequent processes must all be identified through systematic coverage of business process flows. be taken into account for testing. Additional tests are found and the usefulness of the existing tests is assessed prior to functional testing being finished.

**System Test**

System testing guarantees that all requirements are met by the integrated software system as a whole. In order to guarantee known and predictable outcomes, it tests a configuration. The configuration-oriented system integration test is one type of system testing. The foundation of system testing is on process flows and descriptions, with a focus on integration points and pre-driven process links.

**White Box Testing**

White box testing is when the software tester is familiar with the inner workings, structure, and language of the program, or at least its intended use. It has a purpose. It is employed to test regions inaccessible from a black box level.

**Black Box Testing**

Testing software without being aware of the inner workings, structure, or language of the module being tested is known as "black box" testing. Like the majority of other test types, black box tests need to be written from a definitive source document, like a requirements or specification document. It's a test in which treats the software being tested as a "black box." You can't "see" into it. Without taking into account how the software functions, the test gives inputs and reacts to outputs.

**Alpha Testing**

Alpha testing is a combination of three testing techniques: system, integration, and unit testing. The development and quality assurance teams carry out alpha testing. During alpha testing,

The following aspects of the application are examined:

- Spelling Mistakes
- Broken Links
- Cloudy Directions
- Measuring the loading time in the minimum system specifications for optimization.

**Beta Testing**

The product development team proceeds to beta testing after the successful completion of alpha testing. A prerelease version of the program is made available to a sample of users online during beta testing. These early adopters will utilize the program and offer a prompt comments regarding the functionality of the software.

In this testing, the following procedures are carried out:

- Users must install the software and provide the development team with feedback on a regular basis.
- Recording the visual errors and creating crash reports
- The development team addresses the issues after receiving feedback, and a final version is made available to users once all known bugs have been fixed.
- Customer satisfaction is determined by the software's nature. Therefore, as more and more problems are resolved, the product's nature expands, leading to greater customer satisfaction.

**Random Forest Classifier Training and Testing Data**

**Accuracy (Training)**

- The accuracy on the training data is 99.54%, which means the model is able to correctly classify or predict approximately 99 out of every 100 cases in the training dataset.
- High training accuracy like this indicates that the model has learned patterns in the training data very well. Error (Training)

**Error (Training)**

- The error rate is 0.46%, showing that only about 0 or 1 out of every 100 predictions on the training set are incorrect.

- This low error rate complements the high accuracy and further supports the interpretation that the model performs very well on the training data.

**Interpretation**

- "Accuracy (training)" quantifies how often the model makes a correct prediction when evaluated on the training data.
- "Error (training)" is the complement, showing the proportion of wrong predictions.
- Such a high training accuracy often warrants caution: it can indicate overfitting if the test accuracy is much lower, meaning the model may not generalize well to new, unseen data.

**Table 5: Training Data of Accuracy**

| MATRIC | VALUE | INTERPRETATION |
|---|---|---|
| Accuracy(Training) | 99.54 % | Model correctly predicts~99 out of 100 training cases |
| Error(Training) | 0.46% | About 0 to 1out of 100 training prediction are incorrect |
| Accuracy on Test Data: 0.9954545454545455 | | |

**Accuracy (Testing)**

- The testing accuracy is 83.93%, meaning the model correctly predicts about 84 out of every 100 examples it did not see during training.
- This gives a realistic estimate of how well the model will perform in real-world scenarios on new data.

**Error (Testing)**

- The error rate on the test data is 16.07%, which means that about 16 out of every 100 predictions the model makes on unseen data are incorrect.
- A higher error rate compared to the training error suggests the model doesn't perform as well on new data, possibly pointing to overfitting.

**Interpretation**

- Accuracy (testing) shows how often the model is correct on unseen data—this is the key metric to judge how well the model will generalize.
- Error (testing) is the percentage of incorrect predictions on new data.
- The drop in accuracy from training to test data (from about 99.5% to 83.9%) highlights a gap between how well the model fits known data versus new data.

**Table6: Testing of Accuracy**

| MATRIC | VALUE | INTERPRETATION |
|---|---|---|
| Accuracy(Testing) | 83. 93% | Model correctly predicts~84 out of 100 unseen cases |
| Error(Testing) | 16.07% | About 16 out of 100 training prediction are incorrect |
| Accuracy on Test Data: 0.8392857142857143 | | |

**Support Vector Classifier Training and Testing Data**

**Accuracy (Training)**

- Accuracy (86.36%) means the model's predictions are correct for about 86 out of every 100 training samples.
- This indicates good learning performance.

**Error (Training)**

- Error rate (13.64%) is the proportion of incorrect predictions — about 14 out of every 100 training examples are misclassified.

**Interpretation**

- The model shows good performance on the training data, with an accuracy of 86.36%**,** meaning it correctly predicts most cases.
- The error rate of 13.64% indicates that some predictions are still incorrect, suggesting the model has learned well but could be further improved to reduce mistakes.

**Table 7: Training Data of Accuracy**

| MATRIC | VALUE | INTERPRETATION |
|---|---|---|
| Accuracy(Training) | 86.36% | Model correctly predicts~86 out of 100 training cases |
| Error(Training) | 13.64% | About 14 out of 100 training prediction are incorrect |
| Accuracy on Training Data: 0.8636363636363636 | | |

**Accuracy (Testing)**

- Accuracy of 85.71% means the model's predictions on the test data are correct for about 86 out of every 100 cases.

**Error (Testing)**

- Error rate of 14.29% means around 14 predictions out of 100 are wrong, representing the proportion of misclassified cases.

**Interpretation**

- The model has an accuracy of 85.71%, meaning it can correctly predict most of the test cases, showing good overall performance.
- The error rate of 14.29% indicates that the model misclassifies only a small portion of the data, suggesting it is reasonably reliable but still has room for improvement.

**Table8: Testing Data of Accuracy**

| MATRIC | VALUE | INTERPRETATION |
|---|---|---|
| Accuracy(Testing) | 85.71% | Model correctly predicts~86 out of 100 unseen(test) cases |
| Error(Testing) | 14.29% | About 14 out of 100 training prediction are incorrect |
| Accuracy on Test Data: 0.8571428571428571 | | |

# CHAPTER 8
# IMPLEMENTATION

## 8.1 ALGORITHMS

System implementation is a critical phase of the system development life cycle where the designed system is realized as an operational, executable solution within the organization. It involves transforming the approved system design into working software and hardware, ensuring it meets business requirements and is ready for use.

The main goals of system implementation are to complete any remaining design details, write and test programs, prepare user documentation, and train personnel to operate the new system effectively. Implementation also includes deploying the system into the production environment, ensuring data accuracy, validating business functions, and transitioning responsibility from the project team to the operational support team.

There are several approaches to system implementation. The parallel approach runs old and new systems simultaneously for comparison and control, minimizing risks. Other approaches include direct cutover, pilot implementation, and phased implementation, selected based on complexity and risk tolerance.

Training is a key component, including training system operators on routine and exceptional tasks, and end users to operate and troubleshoot the system. Proper training ensures users can adopt the new system smoothly, contributing to overall success.

System installation involves configuring new hardware and software, integrating application software and databases, and performing system validation. Post-installation activities include ongoing support, maintenance, and possibly retiring legacy systems.

Effective system implementation requires careful planning, coordination, and management to minimize disruption and ensure the system meets quality standards and user expectations. It marks the transition from development to operational use, shaping the system's impact on organizational processes and performance.

**Key components of system implementation include:**

- Coding: Translating system designs into actual code.
- Testing: Verifying that code works correctly individually and as a whole system.
- Installation: Deploying the system into the production environment.
- Documentation: Creating manuals and guides for users and maintainers.
- Training: Educating users on how to use the new system effectively.

- Support and Maintenance: Providing ongoing assistance and updates after deployment.

This phase is critical as it moves a system from concept to practice and ensures the system is functional, reliable, and accepted by users.

**8.2 PSEUDO CODING**

```
import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

%matplotlib inline

import seaborn as sn

from sklearn.model_selection import train_test_split

from sklearn.preprocessing import LabelEncoder

from sklearn.metrics import

accuracy_score,classification_report,confusion_matrix,ConfusionMatrixDisplay,roc_curve

lung_cancer = pd.read_csv("/content/lung cancer.csv")

from sklearn.ensemble import RandomForestClassifier

from sklearn.ensemble import HistGradientBoostingClassifier

from sklearn.model_selection import train_test_split

train , test =train_test_split(lung_cancer, test_size=0.2)

print(lung_cancer.GENDER.value_counts())

lung_cancer.replace({"GENDER": {"M": 0, "F": 1}}, inplace = True)

pd

print(lung_cancer.LUNG_CANCER.value_counts())

from sklearn.model_selection import train_test_split
```

```
feature_col_names=["GENDER","AGE","SMOKING","YELLOW_FINGERS","ANXIETY
","PEER_PRESSURE","CHRONICDISEASE","WHEEZING","ALCOHOL
CONSUMING","SHORTNESS OF BREATH","SWALLOWING DIFFICULTY","CHEST
PAIN"]
predicted_class_names=["LUNG_CANCER"]
X=lung_cancer[feature_col_names].values
y=lung_cancer[predicted_class_names].values
split_test_size=0.2
X_train, X_test, Y_train, Y_test = train_test_split(X,y, test_size = 0.2, random_state = 3,
stratify = y)
model=RandomForestClassifier(n_estimators=100,random_state=42)
model.fit(X_train,Y_train)
y_pred=model.predict(X_test)
print("Accuracy:",accuracy_score(Y_test,y_pred))
print(classification_report(Y_test,y_pred))
print(confusion_matrix(Y_test,y_pred))
```

**8.2.1 CODING**

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline
import seaborn as sn
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
```

```
from sklearn.metrics import
accuracy_score,classification_report,confusion_matrix,ConfusionMatrixDisplay,roc_curve
lung_cancer = pd.read_csv("/content/lung cancer.csv")
from sklearn.svm import SVC
from sklearn.model_selection import train_test_split
train , test =train_test_split(lung_cancer, test_size=0.2)
print(lung_cancer.GENDER.value_counts())
lung_cancer.replace({"GENDER": {"M": 0, "F": 1}}, inplace = True)
print(lung_cancer.LUNG_CANCER.value_counts())
from sklearn.model_selection import train_test_split
feature_col_names=["GENDER","AGE","SMOKING","YELLOW_FINGERS","ANXIETY
","PEER_PRESSURE","CHRONIC DISEASE","WHEEZING","ALCOHOL
CONSUMING","SHORTNESS OF BREATH","SWALLOWING DIFFICULTY","CHEST
PAIN"]
predicted_class_names=["LUNG_CANCER"]
X=lung_cancer[feature_col_names].values
y=lung_cancer[predicted_class_names].values
split_test_size=0.2
X_train, X_test, Y_train, Y_test = train_test_split(X,y, test_size = 0.2, random_state = 3,
stratify = y)
from sklearn.metrics import accuracy_score
# 1. Predict on training data
y_train_pred = model.predict(X_train)
train_acc = accuracy_score(Y_train, y_train_pred)
# 2. Predict on test data
```

```
y_test_pred = model.predict(X_test)

test_acc = accuracy_score(Y_test, y_test_pred)

# 3. Print results in percentage and raw

print("Training Accuracy: {:.2f}%".format(train_acc * 100))

print("Test Accuracy: {:.2f}%".format(test_acc * 100))

print("Accuracy (raw):", train_acc, "(train),", test_acc, "(test)")

# 4. Interpretation (like your table)

print(f"The model correctly predicts ~{int(train_acc*100)} out of 100 training samples.")

print(f"The model correctly predicts ~{int(test_acc*100)} out of 100 test samples.")
```

# CHAPTER 9
# PERFORMANCE ANALYSIS

The lung cancer prediction model accurately detects the presence or absence of lung cancer in roughly 87 out of every 100 cases it analyzes, according to the metric "Accuracy" of 87.10%. Because it can distinguish between cancer and non-cancer cases in the great majority of cases, this high accuracy shows that the model is dependable in its overall classification performance. It shows how well the model is able to extract pertinent patterns from the data, which may include imaging inputs, clinical characteristics, or patient history. Since it can assist in the early detection and reasonably confident diagnosis of lung cancer, this level of accuracy holds promise for supporting clinical decisions.

$$\text{Accuracy} = \frac{correct\ predictions}{Total\ predictions} = 0.8709677419354839 \approx 87.10\%$$

However, the model misclassifies roughly 13 cases out of every 100 predictions, as indicated by the error rate of 12.90%. These misclassifications may be false negatives, in which the model fails to detect lung cancer in an affected patient, or false positives, in which the model predicts lung cancer incorrectly for a healthy patient. The clinical impact of these errors should be carefully considered, even though the overall error is relatively low. While false positives could result in needless testing and anxiety, false negatives could postpone necessary treatment.

**Table9: Matrix from the confusion matrix**

| MERTIC | VALUE | INTERPRETATION |
|---|---|---|
| Accuracy | 87.10% | The model correctly predicts lung cancer status in ~87 out of 100c cases |
| Error Rate | 12.90% | Around 12 out of 100 predictions are in correct |

**Precision:**
Precision quantifies the proportion of cases with lung cancer that are actually diagnosed. When the model states "lung cancer," it is typically accurate if precision is high. The formula for precision is:

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{FalsePositives}}$$

The model avoids many false alarms (predicting lung cancer when the patient does not have it) when its precision is high.

**Recall:**

Recall, also known as sensitivity, quantifies the proportion of real-world lung cancer cases that the model accurately detects. The model detects the majority of people who actually have lung cancer if recall is high. The formula for recall is:

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

A high recall indicates that a real cancer case is rarely missed by the model.

# CHAPTER 10
# CONCLUSION AND FUTURE ENHANCENENT

The goal of the project is to use digital image processing and machine learning to enhance early lung cancer diagnosis. It focuses on identifying the various forms of lung cancer, particularly the subtypes of small cell lung cancer (SCLC) and non-small cell lung cancer (NSCLC). The study examines patient data, including symptoms like chest pain and shortness of breath and risk factors like smoking, which are important markers. Because of their capacity to handle intricate medical data, two machine learning algorithms—Random Forest Classifier and HistGradient Boosting Classifier—are employed. With an error rate of 12.90% and an accuracy of 87.10%, the Random Forest model was able to accurately predict lung cancer in roughly 87 out of 100 cases. The model is more effective at identifying cancer cases than non-cancer cases, though, as evidenced by the latter's lower recall and F1-scores due to data imbalance. This study emphasizes the need to address class imbalance in order to improve diagnostic performance, while also highlighting the potential and difficulties of applying machine learning to the detection of lung cancer.

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# PAPER OF PUBLICATIONS

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Certificate of Participation

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He/ She has Published/Presented the paper titled

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Dr.S.Kavitha, M.Sc., M.Phil., Ph.D., Convenor | Dr.R.Thenmozhi, M.Sc., M.Phil., Ph.D., DTP., Principal | Dr.K.Vembanan, M.B.B.S., M.S., Chairman

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